

GCI on Loch — preliminary results

A 36-window titration of a CRY1/KL101 hydration site

Summary

A full 36-window Grand Canonical Integration titration was run end to end on the Loch implementation. Every window completed and passed its checkpoint. The titration curve has the expected sigmoidal shape, the free energy has a well-defined minimum, and the optimal occupancy agrees with the independently computed equilibrium occupancy.

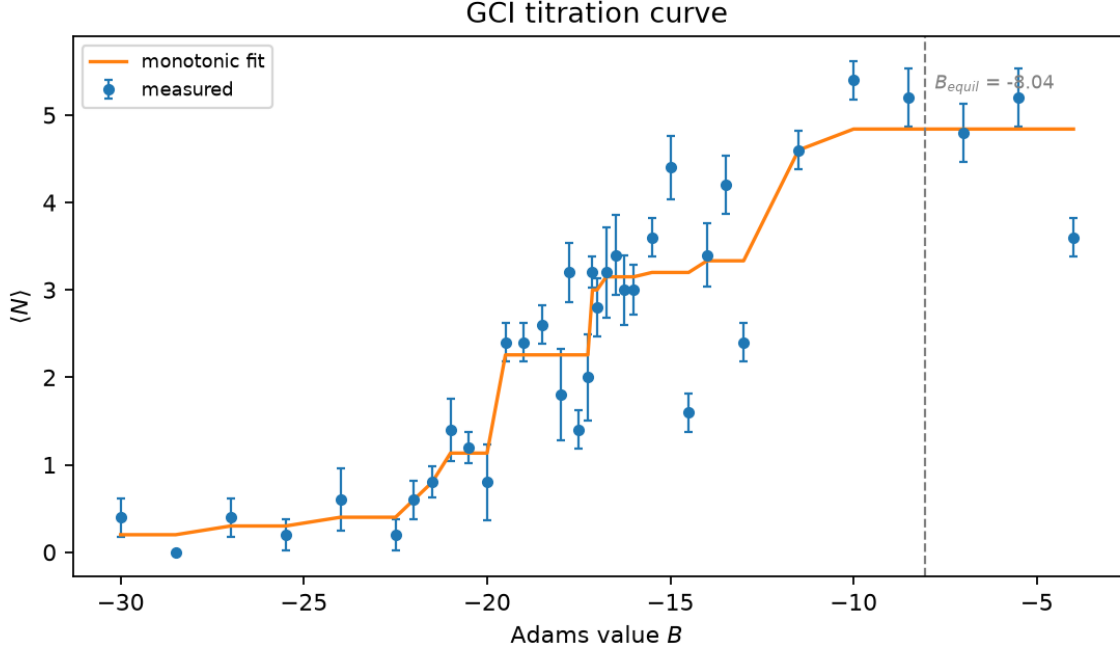
These numbers are not yet publishable. The sweep ran on the short development schedule — 4000 GCMC attempts per window against a production 250 000 — so the curve is noisy and required a monotonic fit before it could be integrated. The purpose here is to show the method and the code path work, not to report a free energy.

1 What was run

System	CRY1/KL101, equilibrated through μ VT1 $\rightarrow$ NPT $\rightarrow$ μ VT2
Site	a pocket water, 2.75 Å from the ligand, 3.09 Å from protein
Sphere	$r = 4$ Å at (48.331, 57.025, 56.001); 27 ghost waters
Ladder	36 values of B from -30 to -4 , finely spaced through the transition
Per window	10 cycles $\times$ (400 GCMC attempts + 200 MD steps) = 4000 attempts, 4 ps
Production would be	625 cycles $\times$ (400 + 4000) = 250 000 attempts, 5 ns

All 36 windows completed, and all 36 checkpoints re-validated (every output re-hashed) when the analysis read them.

2 The titration curve



Occupancy rises from zero at $B = -30$ to a plateau of ~ 5 waters above $B \approx -10$, with the transition between roughly $B = -21$ and $B = -13$. That is the shape a titration must have, and it is not built in anywhere: each point is an independent simulation that only knows its own chemical potential.

Two features are worth pointing at:

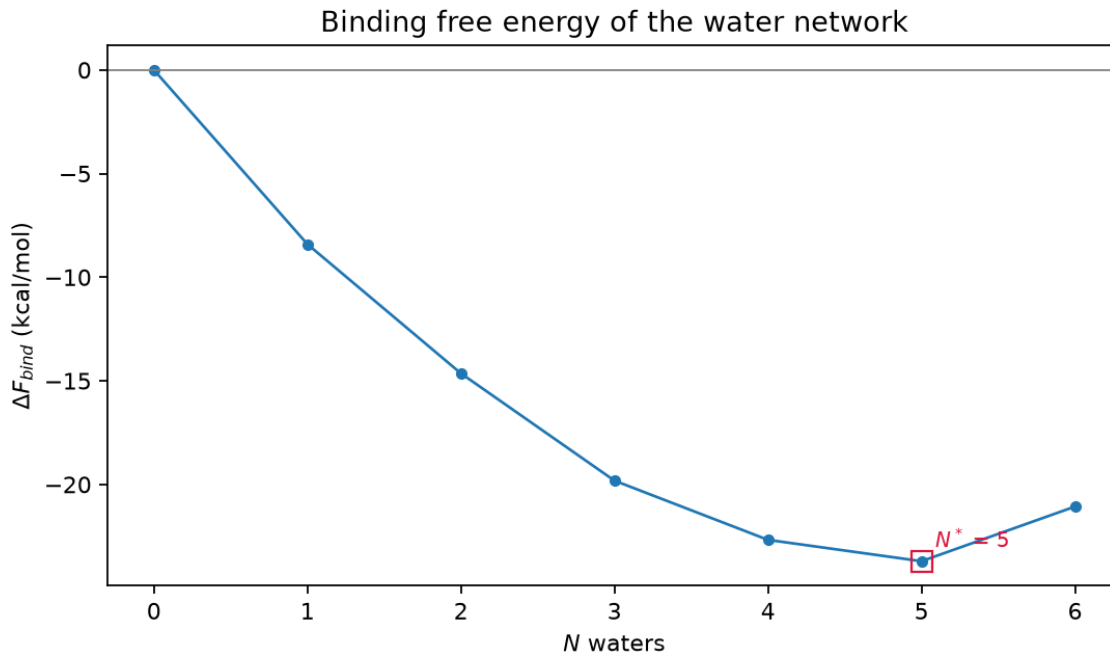
- **The low- B tail reaches zero and stays there.** This matters mechanically, not just aesthetically: it is what sets the lower limit of the integral.
- $B_{\text{equil}} = -8.04$ **falls in the plateau** (dashed line). The region is predicted to be hydrated at equilibrium with bulk water, which is what one expects of a site that was chosen because it contained a water.

The error bars are the spread of the last five checkpoints. They are large relative to the differences between neighbouring windows, which is the direct cause of the problem in §4.

3 The free energy

Integrating the curve by Ross *et al.* (2015) eq. 19,

$$\Delta F_{\text{bind}}(N) = k_B T \left[N B_N - \ln N! - \int_{-\infty}^{B_N} \langle N(B) \rangle dB \right] - N \mu'_{\text{hyd}},$$



N	ΔF_{bind} (kcal/mol)	
0	0.000	(by convention)
1	-8.409	
2	-14.643	
3	-19.818	
4	-22.675	
5	-23.705	← N^* , extrapolated
6	-21.068	extrapolated

The curve turns over cleanly: each added water is worth progressively less until the fifth, after which the network is destabilised. That turnover is the whole point of GCI, and it is present.

Two internal consistency checks pass

- $\langle N \rangle$ at B_{equil} is **4.84**, and the free-energy minimum is at $N^* = \mathbf{5}$. These are computed by different routes — one interpolates the curve, the other integrates it — so their agreement is a real check on the machinery.
- The ghost reservoir never fell below 26 of 27 in any window, so the high- B tail was not clipped by the buffer. This was a live concern: the original study's larger sphere had a buffer roughly equal to its own occupancy.

For orientation only: this gives -4.74 kcal/mol per water, against -4.70 from the published CRY1/AN139 Sphere 2 result ($N^* = 2$, $\Delta F = -9.397$). The agreement is closer than the data deserves — different ligand, different site, under-converged sampling — so treat it as "the right order of magnitude", not as validation.

4 What is not yet right

The curve is not monotonic, and had to be fitted

$\langle N \rangle$ must be non-decreasing in B for the inverse interpolation that gives B_N to mean anything. The measured curve has **14 inversions** — e.g. $\langle N \rangle(-15.0) = 4.4$ but $\langle N \rangle(-14.5) = 1.6$.

These are noise, not physics. Each window is 4 000 attempts and 4 ps of MD, and the tail average is five checkpoints, so the per-point spread (0.4–1.2 waters) is comparable to the spacing between windows.

`gci_analyse.py` refuses by default on such input and names the offending windows. The numbers above come from the opt-in isotonic (monotone least-squares) fit, and are labelled preliminary throughout.

Two further consequences of the short schedule:

- $N^* = 5$ sits at the edge of the sampled range, so B_N for $N \geq 5$ is *extrapolated*. The reported minimum is therefore the least reliable point on the curve.
- The high- B end does not cleanly plateau in the raw data (it wanders between 3.6 and 5.4), which is what pushes the integral onto that extrapolation.

5 Conclusion and next step

The implementation works. A site is anchored by a massless dummy atom, each window samples at a verified Adams value, occupancy is recorded as a machine-readable observable, 36 windows integrate into a free-energy curve with a physically sensible minimum, and two independent consistency checks agree. Nothing in the chain from equilibrated structure to ΔF_{bind} is missing.

What remains is sampling. Re-running with `SMOKE=0` gives $62\times$ more GCMC attempts and $1\,000\times$ more MD per window, which is what the monotonicity and the high- B plateau require. At that point the isotonic fit should become unnecessary — and the right way to report the result is to run the analysis in its default mode and let it either accept the curve or refuse it.

Artifacts: `gci_runs/` (36 windows, each with its own titration CSV, metadata JSON and checkpoint); `figures/` (pooled curve, free-energy table, figures, summary JSON, checkpoint).